

pfSense Firewall Installation and Initial Configuration Guide

This document explains the step-by-step implementation of pfSense Firewall installation and initial LAN configuration using a virtual machine environment. The screenshots below demonstrate interface configuration, LAN setup, DHCP configuration, web access setup, and administrator password configuration.

Step 1

pfSense firewall has successfully booted. The console menu displays WAN and LAN interface details along with available configuration options.

```
pfSense 2.8.0-RELEASE amd64 20250521-2312
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 5206a166fc93a6fc8cb5

*** Welcome to pfSense 2.8.0-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.105/24
                  v6/DHCP6: 2001:2042:6e38:8900::f70/128
LAN (lan) -> em1 -> v4: 192.168.1.1/24
                  v6/t6: 2001:2042:6e38:8904:20c:29ff:fef8:298d/62

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults       13) Update from console
5) Reboot system                  14) Enable Secure Shell (sshd)
6) Halt system                    15) Restore recent configuration
7) Ping host                      16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Step 2

Option 2 (Set Interface IP Address) is selected to configure the LAN interface IP settings.

```
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Bootup complete

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6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 2
```

Step 3

The available interfaces are displayed. LAN interface is selected for IP address configuration.

```
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WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.105/24
                  v6/DHCP6: 2001:2042:6e38:8900::f70/128
LAN (lan) -> em1 -> v4: 192.168.1.1/24
                  v6/t6: 2001:2042:6e38:8904:20c:29ff:fef8:298d/62

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6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 
```

Step 4

Static IPv4 configuration is chosen for the LAN interface, and the new LAN IP address 10.1.1.1 is entered.

```
v6/t6: 2001:2042:6e38:8904:20c:29ff:fef8:298d/62

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 10.1.1.1
```

Step 5

The subnet mask is configured as /24 for the LAN network. IPv6 configuration is skipped.

```
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 10.1.1.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
> n
```

Step 6

DHCP server is enabled on the LAN interface and the IP range from 10.1.1.100 to 10.1.1.199 is configured.

```
Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 10.1.1.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 10.1.1.100
Enter the end address of the IPv4 client address range: 10.1.1.199
```


Step 7

The pfSense system applies the changes and confirms that the LAN interface IP has been changed to 10.1.1.1.

```
Configure IPv6 address LAN interface via DHCP6? (y/n) n
Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 10.1.1.100
Enter the end address of the IPv4 client address range: 10.1.1.199
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) y

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...
Restarting webConfigurator...

The IPv4 LAN address has been set to 10.1.1.1/24
You can now access the webConfigurator by opening the following URL in your web
browser:
    http://10.1.1.1/

Press <ENTER> to continue.█
```

Step 8

The pfSense console now displays the updated LAN IP address configuration.

The IPv4 LAN address has been set to 10.1.1.1/24
You can now access the webConfigurator by opening the following URL in your web browser:

http://10.1.1.1/

Press <ENTER> to continue.

VMware Virtual Machine - Netgate Device ID: 5206a166fc93a6fc8cb5

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v6/DHCP6: 2001:2042:6e38:8900::f70/128

LAN (lan) -> em1 -> v4: 10.1.1.1/24

- | | |
|-------------------------------------|----------------------------------|
| 0) Logout / Disconnect SSH | 9) pfTop |
| 1) Assign Interfaces | 10) Filter Logs |
| 2) Set interface(s) IP address | 11) Restart GUI |
| 3) Reset admin account and password | 12) PHP shell + pfSense tools |
| 4) Reset to factory defaults | 13) Update from console |
| 5) Reboot system | 14) Enable Secure Shell (sshd) |
| 6) Halt system | 15) Restore recent configuration |
| 7) Ping host | 16) Restart PHP-FPM |
| 8) Shell | |

Enter an option: 3

Step 9

Option 3 is selected to reset and configure the administrator account password.

Press <ENTER> to continue.

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*** Welcome to pfSense 2.8.0-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.105/24
v6/DHCP6: 2001:2042:6e38:8900::f70/128

LAN (lan) -> em1 -> v4: 10.1.1.1/24

- | | |
|-------------------------------------|----------------------------------|
| 0) Logout / Disconnect SSH | 9) pfTop |
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| 8) Shell | |

Enter an option: 3

The authentication configuration and privileges for the "admin" account will be reset to the default.

Proceed? (y/n): y

Step 10

The system confirms that the administrator account will be reset to default settings before assigning a new password.

```
0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults       13) Update from console
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6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell
```

Enter an option: 3

The authentication configuration and privileges for the "admin" account will be reset to the default.

Proceed? (y|n): y

The default administrator account in the User Manager ("admin") has been reset. The password must now be set to a new value.

Changing password for 'admin' in the User Manager database.

Hints:

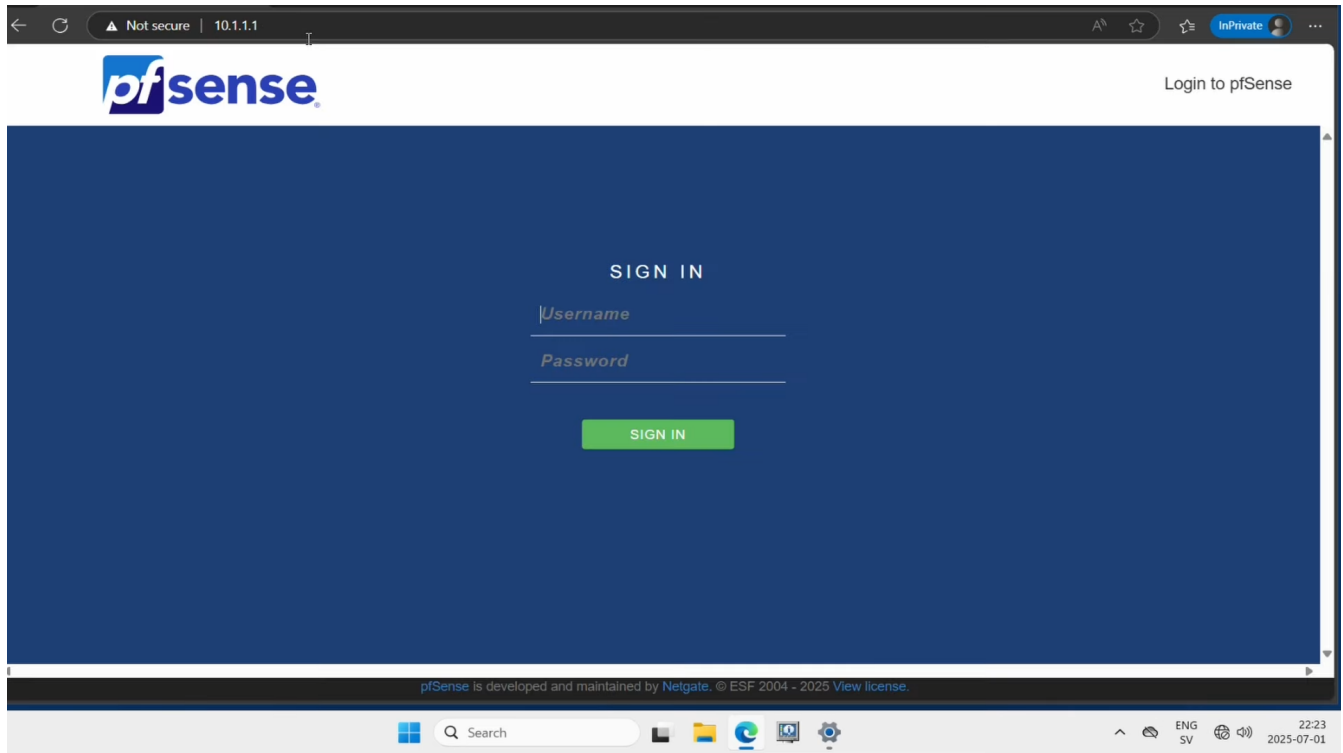
* Current NIST guidelines prioritize password length over complexity.

* The password cannot be identical to the username.

New Password: █

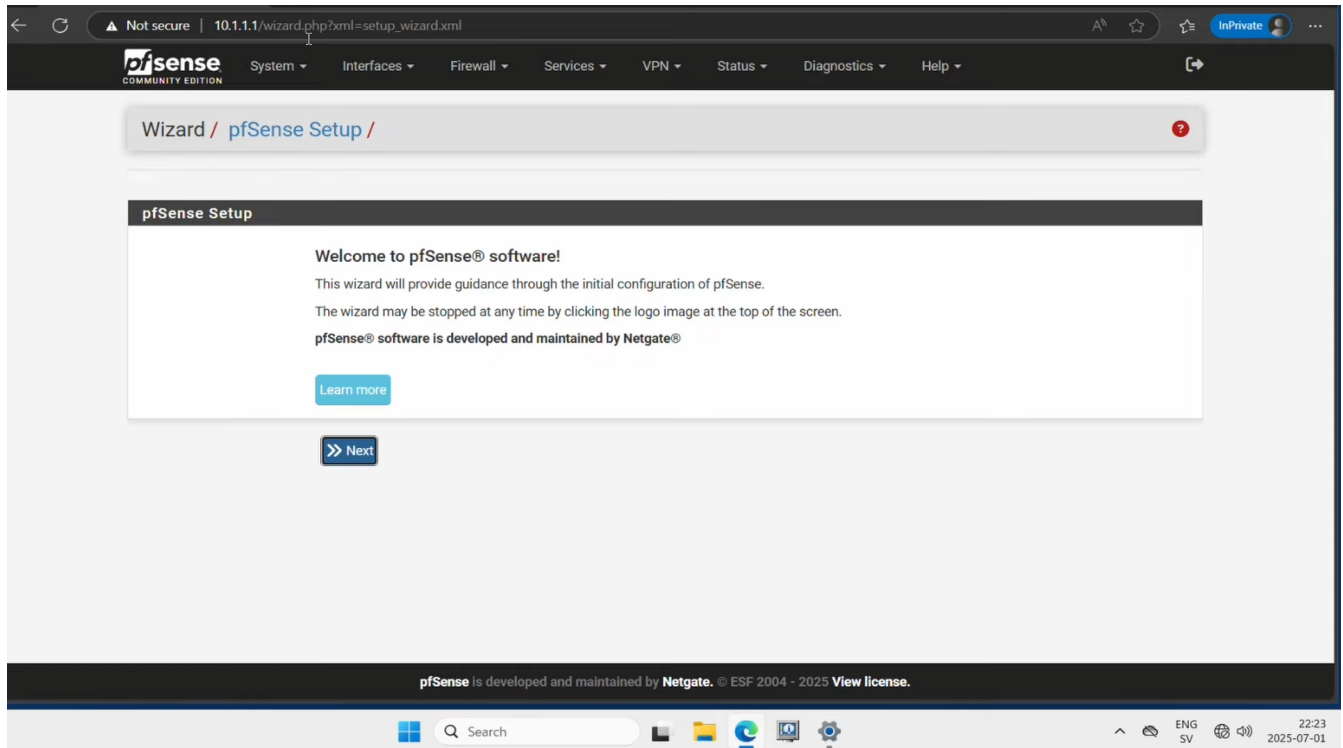
Step 11

A new password is configured for the pfSense administrator account.



Step 12

The pfSense web configurator login page is accessed using the configured LAN IP address 10.1.1.1.



Step 13

The pfSense setup wizard is opened in the web configurator to continue the initial setup process.

The screenshot shows the pfSense web configurator interface. At the top is a dark navigation bar with the pfSense logo and menu items: System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. Below this is a breadcrumb trail: Wizard / pfSense Setup / Configure LAN Interface. A red progress bar indicates 'Step 5 of 9'. The main section is titled 'Configure LAN Interface' and contains the instruction: 'On this screen the Local Area Network information will be configured.' There are two input fields: 'LAN IP Address' with the value '10.1.1.1' and a note 'Type dhcp if this interface uses DHCP to obtain its IP address.', and 'Subnet Mask' with a dropdown menu showing '24'. A blue button with a double arrow and the text 'Next' is positioned below the fields. At the bottom of the page, a footer states: 'pfSense is developed and maintained by Netgate. © ESF 2004 - 2025 View license.'

Step 14

The LAN interface settings are verified in the pfSense web configuration wizard, including LAN IP address and subnet mask.

System ▾Interfaces ▾Firewall ▾Services ▾VPN ▾Status ▾Diagnostics ▾Help ▾

Wizard / pfSense Setup / Change admin Account Password

Step 6 of 9

Change admin Account Password

Change the password for the admin account.
This account is used to access the GUI, console (if protected), and SSH service (if enabled).

New admin Password

Confirm admin Password

>> Next

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Conclusion:

The pfSense firewall installation and initial configuration were completed successfully. The LAN network was configured with a static IP address, DHCP services were enabled, and administrator access was secured with a new password. The firewall can now be managed through the pfSense web configurator interface.